

Anika E. Chawla

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

M.S. Aerospace Engineering

August 2025 - (Expected) May 2026

B.S. Aerospace Engineering, Minor in Computing & Intelligence - GPA: 3.76

August 2021 - May 2025

WORK EXPERIENCE

Pratt & Whitney (RTX) — AeroThermal Fluids Intern, Advanced Concepts

May 2024 – August 2024

- Developed AI-based analytics tools to identify linear, limiting, and clustering trends in large engine-architecture datasets, revealing key design sensitivities influencing engine performance
- Applied XGBoosted trees, correlation analysis, and DBSCAN clustering to explore high-dimensional architecture trade spaces and applied XAI methods such as SHAP to analyze design decision impact
- Conducted techno-economic assessments of hydrogen fuel and sustainable aviation fuel (SAF) pathways for next-generation propulsion concepts

MITRE — Modeling & Simulation Intern, Cross Cutting Urgent Innovation Cell

May 2023 – August 2023

- Developed routing algorithms for maritime environments under uncertainty using Dijkstra's algorithm, Monte Carlo Tree Search, and genetic algorithms
- Modeled satellite orbit propagation and SAR/EOIR sensor projections to assess LEO ground coverage under constraints
- Validated algorithm outputs through unit testing and comparison with AFSIM physics-based simulations

RESEARCH EXPERIENCE

FAA CLEEN ASCENT Project, Aerospace Systems Design Laboratory

Graduate Research Assistant

Fall 2025 – Present

- Evaluated technology readiness and integration pathways, assessing how adoption timelines and system compatibility affect long-term sustainability outcomes
- Conducted engine cycle and system-level analyses using EDS, FLOPS, and Python to quantify fuel burn, noise, and emissions under varying modeling and policy assumptions
- Performed fleet-level scenario analyses incorporating aircraft turnover and heterogeneous technology adoption to assess robustness of environmental outcomes

Wireless Energy from Beamed Signals CubeSat Development, Space Systems Design Laboratory

Undergraduate Research Assistant

Fall 2024

- Optimized satellite imagery edge-detection and feature-extraction algorithms from IR camera outputs using OpenCV, affine transformations, and moment invariants to maintain robustness under image distortion
- Developed image reconstruction and validation algorithms using flux and streak analysis, benchmarking against known starfields to verify spacecraft orientation accuracy

Model-Based Systems Engineering, Aerospace Systems Design Laboratory

Undergraduate Research Assistant

Fall 2022 – Spring 2024

- Modeled aerospace system lifecycle architectures using SysML to support traceability, impact analysis, and system trade studies
- Developed propulsion-system design tools for early-stage architectural decisions and validated outputs using flow characterization

TECHNICAL PROJECTS

Digital Services & MRO — Airbus Grand Challenge

Fall 2025 – Present

- Prototyped an explainability layer for predictive maintenance alerts on Skywise platform using representative deep learning and explainable models on C-MAPSS and Hawk T1A datasets
- Applied SHAP, LIME, Integrated Gradients, and other XAI tools to diagnose alert drivers, failure precursors, and model confidence

HARMONY Mars Mission — Aerospace Engineering Capstone

Fall 2024 – Spring 2025

- Led systems engineering for preliminary CubeSat architecture design, including requirements development, interface control, and subsystem integration
- Synthesized subsystem analyses into mission-level architectures, resolving cross-coupled constraints and developing mass, power, and resource budgets

SKILLS

Programming: Python, C, Java, Git

Systems & Simulation: MATLAB, NPSS, EDS, FLOPS, AFSIM, SysML (MagicDraw)